CURRICULUM COMPANION : TECHNICAL LEARNING MODULES

Week 1 Learning Materials : Foundations & Data Engineering

Core Concepts, Mathematical Formulations, Defensive Ingestion & Leak-Free Pipelines

TARGET AUDIENCE	ESTIMATED READING TIME	CORE PREREQUISITES	PRIMARY OBJECTIVE
ML Interns (Trial Phase)	35-45 Minutes	Python 3.10+, Git, Pandas	Leak-Free Pipeline Discipline

01 Virtual Environment Isolation & Atomic Git Hygiene

MODULE 1

Isolated environments prevent package conflicts between client projects. Pin exact versions in **`requirements.txt`** to ensure zero production drift. Commit code atomically following Conventional Commits format (`feat:`, `fix:`, `test:`, `docs:`). Never commit large raw data or virtual environment artifacts to version control.

```
# Virtualenv initialization and dependency management
$ python -m venv .venv && source .venv/bin/activate # On Windows: .\.venv\Scripts\Activate.ps1
$ pip install -r requirements.txt && pytest -v && ruff check .
```

02 Defensive Data Ingestion & Health Diagnostics

MODULE 2

Real-world telemetry data arrives with missing timestamps, corrupt rows, and unexpected encodings. Your loader must fail fast on invalid files and compute memory diagnostics before running training:

DEFENSIVE INGESTION RULEBOOK

1. Defensive Type Casting: Force string columns to strip whitespace; replace empty strings and "nan" tokens with explicit "UNKNOWN" categories.
2. Skew-Aware Imputation: Never use mean imputation on right-skewed operational metrics (such as session duration). Always use median values to resist extreme outliers.
3. Memory Footprint: Downcast 64-bit integers to 32-bit where allowable and convert low-cardinality strings to categorical dtypes to save 60-80% RAM.

03 Strict Data Leakage Safeguards & Stratification

MODULE 3

Data Leakage occurs when information from outside the training dataset is used to create or tune the model. A common rookie mistake is fitting scalers or imputers on the whole dataset before splitting.

THREE UNBREAKABLE RULES OF LEAK-FREE SPLITTING

- ✓ **Split Before Transforming:** Partition data into train (70%), validation (15%), and test (15%) splits before calculating ANY mean, median, IQR, or scaler parameters.
- ✓ **Index Disjointness:** Formally verify with sets that `set(train_idx).isdisjoint(test_idx)` evaluates to True.
- ✓ **Stratified Sampling:** In imbalanced datasets (e.g. 15% fraud), random splitting can yield unrepresentative test subsets. Always stratify across the target label.

04 High-Performance Columnar Storage : Parquet vs CSV

MODULE 4

While CSV stores data row-by-row as uncompressed text, Apache Parquet uses columnar encoding with embedded schema metadata, snappy compression, and column pruning capabilities:

FEATURE	CSV (COMMA-SEPARATED)	APACHE PARQUET (SNAPPY)
Storage Format	Row-oriented plaintext	Column-oriented binary
Disk Footprint	Large (e.g. 2.63 MB raw benchmark)	Compact (approx 0.7 - 0.9 MB with Snappy)
Schema Preservation	None; dtypes re-inferred on each load	Strict embedded schema & column dtypes
Query Performance	Must scan entire file sequentially	Reads only requested columns (column projection)

05 Statistical Hypothesis Testing Frameworks

MODULE 5

Never rely solely on visual impressions from charts. Ground your findings in statistical hypothesis testing:

TEST A Chi-Square Test of Independence (χ^2)

Categorical vs Categorical

Evaluates whether two categorical variables (e.g. Device OS vs Fraud Flag) are statistically independent.

```
# Null Hypothesis (H0): Device OS and Abuse Rate are independent.
>>> chi2, p_val, dof, expected = scipy.stats.chi2_contingency(contingency_table)
# Decision: If p_val < 0.05, reject H0 (statistically significant association exists).
```

TEST B Mann-Whitney U Test (Wilcoxon Rank-Sum)

Continuous vs Binary Label

Non-parametric test comparing the distributions of two independent groups when normality cannot be assumed (e.g. skewed session durations).

```
# Null Hypothesis (H0): Normal and Fraudulent sessions share identical distributions.
>>> stat, p_val = scipy.stats.mannwhitneyu(benign_durations, fraud_durations)
```

06 Unit Testing Data Pipelines & Mentor Interview Prep

MODULE 6

- ☐ **Empty File Handling:** Verify `load_data` raises `ValueError` on 0-byte files.
- ☐ **Zero Data Leakage:** Assert train, val, and test index sets are completely disjoint.
- ☐ **Split Ratio Accuracy:** Verify sum of split lengths equals exactly total row count.
- ☐ **Documentation & Typing:** Every function has complete Google docstrings and type annotations.

WEEKEND 1-ON-1 MENTOR PROBE PREPARATION

During your weekend feedback session, your ML Lead will review your PR and ask: (1) "Why did you choose median over mean imputation for telemetry metrics?", (2) "Prove that your preprocessing pipeline introduces zero data leakage," and (3) "What did your Chi-Square contingency table reveal regarding device abuse?" Be prepared to walk through your code and charts directly.